A First Course in Abstract Algebra

Section 11: Direct Products and Finitely Generated Abelian Groups

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

11 Direct Products and Finitely Generated Abelian Groups 4

Exercise 11.1: Elements and orders in $\mathbb{Z}_2 \times \mathbb{Z}_4$ 4

Exercise 11.2: Elements and orders in $\mathbb{Z}_3 \times \mathbb{Z}_4$ 4

Exercise 11.3: Order of $(2, 6)$ 4

Exercise 11.4: Order of $(2, 3)$ 4

Exercise 11.5: Order of $(8, 10)$ 5

Exercise 11.6: Order of $(3, 10, 9)$ 5

Exercise 11.7: Order of $(3, 6, 12, 16)$ 5

Exercise 11.8: Largest orders of cyclic subgroups 5

Exercise 11.9: Proper nontrivial subgroups of $\mathbb{Z}_2^2$ 5

Exercise 11.10: Proper nontrivial subgroups of $\mathbb{Z}_2^3$ 6

Exercise 11.11: Order-four subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4$ 6

Exercise 11.12: Klein subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4 imes \mathbb{Z}_4$ 6

Exercise 11.13: Direct product descriptions of $\mathbb{Z}_{60}$ 7

Exercise 11.14: Fill in the numerical blanks 7

Exercise 11.15: Maximum order in $\mathbb{Z}_4 imes \mathbb{Z}_6$ 7

Exercise 11.16: Comparing two groups of order 24 . . . 7
Exercise 11.17: Maximum order in a three-factor product . . . 7
Exercise 11.18: Comparing the groups in Exercise 18 . . . 8
Exercise 11.19: Maximum order in $\mathbb{Z}_4 imes \mathbb{Z}_{18} imes \mathbb{Z}_{15}$ . . . 8
Exercise 11.20: Comparing the groups in Exercise 20 . . . 8
Exercise 11.21: Abelian groups of order 8 . . . 8
Exercise 11.22: Abelian groups of order 16 . . . 8
Exercise 11.23: Abelian groups of order 32 . . . 9
Exercise 11.24: Abelian groups of order 720 . . . 9
Exercise 11.25: Abelian groups of order 1089 . . . 9
Exercise 11.26: Counts for orders 24, 25, and 600 . . . 9
Exercise 11.27: Multiplicativity for coprime group orders . . . 10
Exercise 11.28: Groups of order 10^5 . . . 10
Exercise 11.29: Partition counts for prime-power orders . . . 10
Exercise 11.30: A Cayley digraph on a toroidal grid . . . 10
Exercise 11.31: Two concentric polygon digraphs . . . 11
Exercise 11.32: True or false: direct products . . . 11
Exercise 11.33: An indecomposable abelian group . . . 12
Exercise 11.34: Subgroups isomorphic to the full product . . . 12
Exercise 11.35: A composite-order group with no nontrivial direct decomposition . . . 12
Exercise 11.36: True or false: classification of abelian groups . . . 12
Exercise 11.37: Changing the prime in a prime-power order . . . 12
Exercise 11.38: Order-eight and order-four subgroups of an order-72 group . . . 13
Exercise 11.39: The torsion subgroup . . . 13
Exercise 11.40: Orders of two torsion subgroups . . . 13
Exercise 11.41: Torsion in the nonzero real numbers . . . 13
Exercise 11.42: Torsion in the nonzero complex numbers . . . 13
Exercise 11.43: Torsion and free parts of a finitely generated abelian group . . . 13
Exercise 11.44: Torsion coefficients . . . 14
Exercise 11.45: Synopsis of the coprime-product criterion . . . 14
Exercise 11.46: A direct product of abelian groups is abelian . . . 14
Exercise 11.47: Elements of order at most two . . . 14
Exercise 11.48: Identity together with all elements of an exact order . . . 14
Exercise 11.49: Why commutativity is needed for involutions . . . 15

Exercise 11.50: The coordinate subgroups in an external product 15
Exercise 11.51: Recognizing an internal direct product 15
Exercise 11.52: Detecting a noncyclic finite abelian group 15
Exercise 11.53: Element orders in a prime-power group 16
Exercise 11.54: Cancellation for finitely generated abelian groups 16

11 Direct Products and Finitely Generated Abelian Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 11.1: Elements and orders in $\mathbb{Z}_2 \times \mathbb{Z}_4$

The elements are all pairs (a, b) with $0 \le a < 2$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 2$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

$a\backslash b$	0	1	2	3
0	1	4	2	4
1	2	4	2	4

No element has order 8, so the order-8 group is not cyclic.

Exercise 11.2: Elements and orders in $\mathbb{Z}_3 \times \mathbb{Z}_4$

The elements are all pairs (a, b) with $0 \le a < 3$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 3$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

$a\backslash b$	0	1	2	3
0	1	4	2	4
1	3	12	6	12
2	3	12	6	12

The element $(1, 1)$ has order $\operatorname{lcm}(3, 4) = 12$, equal to the group order, so it generates the whole group; the group is cyclic.

Exercise 11.3: Order of $(2, 6)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12}$, the component orders are

$$\frac{4}{\gcd(4, 2)} = 2, \quad \frac{12}{\gcd(12, 6)} = 2.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2, 6) = \operatorname{lcm}(2, 2) = 2.$$

Exercise 11.4: Order of $(2, 3)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_6 \times \mathbb{Z}_{15}$, the component orders are

$$\frac{6}{\gcd(6, 2)} = 3, \quad \frac{15}{\gcd(15, 3)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2,3) = \operatorname{lcm}(3,5) = 15.$$

Exercise 11.5: Order of $(8, 10)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_{12} \times \mathbb{Z}_{18}$, the component orders are

$$\frac{12}{\gcd(12,8)} = 3, \quad \frac{18}{\gcd(18,10)} = 9.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(8,10) = \operatorname{lcm}(3,9) = 9.$$

Exercise 11.6: Order of $(3, 10, 9)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{15}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,10)} = 6, \quad \frac{15}{\gcd(15,9)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,10,9) = \operatorname{lcm}(4,6,5) = 60.$$

Exercise 11.7: Order of $(3, 6, 12, 16)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20} \times \mathbb{Z}_{24}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,6)} = 2, \quad \frac{20}{\gcd(20,12)} = 5, \quad \frac{24}{\gcd(24,16)} = 3.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,6,12,16) = \operatorname{lcm}(4,2,5,3) = 60.$$

Exercise 11.8: Largest orders of cyclic subgroups

In $\mathbb{Z}_m \times \mathbb{Z}_n$, every element order divides $\operatorname{lcm}(m,n)$, since that many additions kill both coordinates. The element $(1,1)$ attains this order, so it is the largest cyclic-subgroup order. Thus the answers are

$$\operatorname{lcm}(6,8) = 24, \qquad \operatorname{lcm}(12,15) = 60.$$

These need not equal the group orders 48 and 180, respectively.

Exercise 11.9: Proper nontrivial subgroups of $\mathbb{Z}_2^2$

All three nonzero elements have order 2. They generate the three subgroups

$$\{(0,0),(1,0)\}, \quad \{(0,0),(0,1)\}, \quad \{(0,0),(1,1)\}.$$

Any subgroup containing two different nonzero vectors contains their sum, the third nonzero vector, and so is the whole group. Thus the three listed subgroups are exactly the proper nontrivial ones.

Exercise 11.10: Proper nontrivial subgroups of $\mathbb{Z}_2^3$

Regard the group as the three-dimensional vector space over $\mathbb{Z}_2$; subgroup closure is exactly linear-subspace closure, because the only scalars are $0, 1$. The seven subgroups of order 2 are $\{0, v\}$ for

$$v = (1,0,0), (0,1,0), (0,0,1), (1,1,0), (1,0,1), (0,1,1), (1,1,1).$$

The seven subgroups of order 4 are the planes described, respectively, by

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y = 0, \quad x + z = 0, \quad y + z = 0, \quad x + y + z = 0,$$

where arithmetic is modulo 2. Each displayed equation has four solutions. Every two-dimensional subspace is the kernel of one nonzero linear functional, and over $\mathbb{Z}_2$ no two distinct nonzero functionals are scalar multiples. Consequently this lists all seven planes. Proper nontrivial subgroups have dimension 1 or 2, so these fourteen are all of them.

Exercise 11.11: Order-four subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4$

An element has order 4 precisely when its second coordinate is odd. There are four such elements, and each cyclic order-4 subgroup has two generators. The two distinct cyclic subgroups are

$$\langle (0,1) \rangle = \{(0,0), (0,1), (0,2), (0,3)\},$$

$$\langle (1,1) \rangle = \{(0,0), (1,1), (0,2), (1,3)\}.$$

Any noncyclic group of order 4 is a Klein group and all its elements are killed by 2. The elements killed by 2 here are exactly

$$\{(0,0), (1,0), (0,2), (1,2)\},$$

which form one more subgroup. This exhausts both possible types of order-4 subgroup.

Exercise 11.12: Klein subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4 imes \mathbb{Z}_4$

Every element of a Klein subgroup is killed by 2, so it must lie in

$$T = \{(a, 2b, 2c) : a, b, c \in \mathbb{Z}_2\} \cong \mathbb{Z}_2^3.$$

A Klein subgroup is a two-dimensional subspace of this three-dimensional space. Thus all seven are given by $(a, 2b, 2c)$ subject to one of the conditions

$$a = 0,\ b = 0,\ c = 0,\ a + b = 0,\ a + c = 0,\ b + c = 0,\ a + b + c = 0$$

in $\mathbb{Z}_2$. For example the first is $\{(0,0,0), (0,2,0), (0,0,2), (0,2,2)\}$. Each condition gives four elements, all of exponent 2, hence a Klein group. The seven-plane classification from the preceding exercise proves completeness.

Exercise 11.13: Direct product descriptions of $\mathbb{Z}_{60}$

A finite direct product of cyclic groups is cyclic precisely when their orders are pairwise relatively prime. Since $60 = 4 \cdot 3 \cdot 5$, the prime-power blocks $4, 3, 5$ must remain intact but can be grouped. Ignoring order of factors, all descriptions using at least two nontrivial factors are

$$\mathbb{Z}_4 \times \mathbb{Z}_{15}, \quad \mathbb{Z}_3 \times \mathbb{Z}_{20}, \quad \mathbb{Z}_5 \times \mathbb{Z}_{12}, \quad \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_5.$$

These exhaust the partitions of the three prime-power blocks into at least two groups. If factors $\mathbb{Z}_1$ are permitted, one can insert arbitrarily many trivial factors, giving infinitely many formally different descriptions without changing the group; the finite list uses the usual convention of omitting trivial factors.

Exercise 11.14: Fill in the numerical blanks

(a) $\operatorname{ord}_{\mathbb{Z}_{24}}(18) = 24/\gcd(24, 18) = 24/6 = 4$.

(b) $|\mathbb{Z}_3 \times \mathbb{Z}_4| = 3 \cdot 4 = 12$.

(c) The orders of $4 \in \mathbb{Z}_{12}$ and $2 \in \mathbb{Z}_8$ are 3 and 4, so $(4, 2)$ has order 12.

(d) $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$; both blanks are 2.

(e) A finite-order element of $\mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}_4$ must have zero middle coordinate. There are $2 \cdot 1 \cdot 4 = 8$ such elements, all indeed of finite order.

Exercise 11.15: Maximum order in $\mathbb{Z}_4 times \mathbb{Z}_6$

Every component is killed by its modulus, so $12 = \operatorname{lcm}(4, 6)$ kills every pair. The pair $(1, 1)$ has component orders $4, 6$ and hence order 12. Thus 12 is the maximum possible order.

Exercise 11.16: Comparing two groups of order 24

Yes. Splitting relatively prime cyclic factors gives

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3, \qquad \mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3.$$

The right-hand products differ only in order of coordinates, so they are isomorphic. An explicit map from the second group to the first is $(a, b) \mapsto (b \bmod 2, c)$, where c is the unique residue modulo 12 satisfying $c \equiv a \pmod 4$, $c \equiv b \pmod 3$; existence and uniqueness follow from coprimality of $4, 3$.

Exercise 11.17: Maximum order in a three-factor product

For $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$ the maximum is $\operatorname{lcm}(8, 10, 24) = 2^3 \cdot 3 \cdot 5 = 120$. Every element is killed by 120, and $(1, 1, 1)$ has exactly that order, proving both the upper bound and its attainment.

Exercise 11.18: Comparing the groups in Exercise 18

The groups are not isomorphic, although both have the same order and the same maximum element order 120. Count elements killed by 4. In $\mathbb{Z}_m$ the congruence $4x = 0$ has $\gcd(4, m)$ solutions. Therefore the counts are

$$\gcd(4, 8)\gcd(4, 10)\gcd(4, 24) = 4 \cdot 2 \cdot 4 = 32,$$

$$\gcd(4, 4)\gcd(4, 12)\gcd(4, 40) = 4 \cdot 4 \cdot 4 = 64.$$

An isomorphism preserves the equation $4x = 0$ and hence its number of solutions, so the different counts rule out an isomorphism.

Exercise 11.19: Maximum order in $\mathbb{Z}_4 imes \mathbb{Z}_{18} imes \mathbb{Z}_{15}$

The least common multiple of the factor orders is $\operatorname{lcm}(4, 18, 15) = 2^2 \cdot 3^2 \cdot 5 = 180$. It kills every element, and $(1, 1, 1)$ has that order. Therefore the maximum is 180.

Exercise 11.20: Comparing the groups in Exercise 20

Yes. Decompose the factors into prime powers:

$$\mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15} \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_3 \times \mathbb{Z}_5,$$
$$\mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10} \cong \mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_2 \times \mathbb{Z}_5.$$

The same five prime-power cyclic factors occur, so reordering coordinates gives an isomorphism. Splitting was legitimate because each original factor was split into relatively prime factors.

Exercise 11.21: Abelian groups of order 8

The classification theorem assigns one group to each partition of 3, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\mathbb{Z}_8,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 3 groups in all.

Exercise 11.22: Abelian groups of order 16

The classification theorem assigns one group to each partition of 4, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\mathbb{Z}_{16},$$
$$\mathbb{Z}_8 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_4,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 5 groups in all.

Exercise 11.23: Abelian groups of order 32

The classification theorem assigns one group to each partition of 5, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\begin{gathered}\mathbb{Z}_{32},\\ \mathbb{Z}_{16}\times\mathbb{Z}_2,\\ \mathbb{Z}_8\times\mathbb{Z}_4,\\ \mathbb{Z}_8\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_4\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2.\end{gathered}$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 7 groups in all.

Exercise 11.24: Abelian groups of order 720

Factor $720 = 2^4 \cdot 3^2 \cdot 5$. The possible 2-primary factors are

$$T \in \{\mathbb{Z}_{16},\ \mathbb{Z}_8\times\mathbb{Z}_2,\ \mathbb{Z}_4\times\mathbb{Z}_4,\ \mathbb{Z}_4\times\mathbb{Z}_2^2,\ \mathbb{Z}_2^4\},$$

and the possible 3-primary factors are $U \in \{\mathbb{Z}_9, \mathbb{Z}_3^2\}$. The 5-primary factor must be $\mathbb{Z}_5$. All groups, with none omitted or repeated up to isomorphism, are therefore

$$\boxed{T\times U\times\mathbb{Z}_5 \quad \text{for every one of the five } T \text{ and two } U.}$$

This specifies $5\cdot 2 = 10$ groups explicitly by independent choices; uniqueness of the prime-power cyclic decomposition distinguishes them.

Exercise 11.25: Abelian groups of order 1089

Since $1089 = 33^2 = 3^2\cdot 11^2$, each primary component has two possibilities. The four groups are

$$\begin{aligned}&\mathbb{Z}_9\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_9\times\mathbb{Z}_{11}^2,\\ &\mathbb{Z}_3^2\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_3^2\times\mathbb{Z}_{11}^2.\end{aligned}$$

The exponent-2 partition choices are 2 and $1+1$ for each prime. They produce all primary decompositions and hence all isomorphism types.

Exercise 11.26: Counts for orders 24, 25, and 600

The printed numbers are 24 and 25, not 2^4 and 2^5. Since $24 = 2^3\cdot 3$, the three partitions of 3 give three groups:

$$\mathbb{Z}_8\times\mathbb{Z}_3, \quad \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_3, \quad \mathbb{Z}_2^3\times\mathbb{Z}_3.$$

For order $25 = 5^2$ the two possibilities are $\mathbb{Z}_{25}$ and $\mathbb{Z}_5^2$. Because 24 and 25 are coprime, their primary decompositions can be chosen independently. Hence for $(24)(25) = 600$ there are $3 \cdot 2 = 6$ abelian groups up to isomorphism.

Exercise 11.27: Multiplicativity for coprime group orders

Suppose $\gcd(m, n) = 1$. The prime factors of m and n are disjoint. By the finite abelian classification theorem, any abelian group of order mn is uniquely determined by its primary factors. Collecting the factors at primes dividing m gives a group of order m; collecting those at primes dividing n gives one of order n. Conversely, the direct product of any one of the r types of order m and any one of the s types of order n gives a group of order mn. Uniqueness of primary decompositions means different pairs give different isomorphism types. Thus there are exactly rs.

Exercise 11.28: Groups of order 10^5

Factor $10^5 = 2^5 \cdot 5^5$. There are seven partitions of 5, hence seven abelian 2-groups of order 2^5 and seven abelian 5-groups of order 5^5. These choices are independent by coprimality, so the answer is $7 \cdot 7 = 49$.

Exercise 11.29: Partition counts for prime-power orders

(a) An abelian group of order p^n corresponds to a partition of n: parts $\lambda_1, \ldots, \lambda_k$ give $\mathbb{Z}_{p^{\lambda_1}} \times \cdots \times \mathbb{Z}_{p^{\lambda_k}}$. The required table is

n	2	3	4	5	6	7	8
number	2	3	5	7	11	15	22

One reproducible way to obtain it is the recurrence $P(n, k) = P(n, k-1) + P(n-k, k)$, splitting partitions according to whether a part k occurs, with $P(0, k) = 1$ and $P(n, 0) = 0$ for $n > 0$; the desired count is $P(n, n)$. **(b)** Distinct primes permit independent choices. Thus

$$\begin{aligned} \text{(i) } & p^3q^4r^7 : 3 \cdot 5 \cdot 15 = 225, \\ \text{(ii) } & (qr)^7 = q^7r^7 : 15 \cdot 15 = 225, \\ \text{(iii) } & q^5r^4q^3 = q^8r^4 : 22 \cdot 5 = 110. \end{aligned}$$

In (iii) the two powers of the same prime q must first be combined; they are not independent primary factors.

Exercise 11.30: A Cayley digraph on a toroidal grid

Place a vertex (i, j) in an m-by-n grid, with $0 \le i < m$, $0 \le j < n$. Draw one solid arrow

$$(i, j) \longrightarrow (i + 1 \bmod m, j)$$

for the generator $(1, 0)$, and one dashed arrow

$$(i, j) \dashrightarrow (i, j + 1 \bmod n)$$

for $(0,1)$. Thus each row wraps into an m-cycle and each column into an n-cycle. A corner of the schematic is

$$\begin{array}{ccccc} (0,2) & \longrightarrow & (1,2) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,1) & \longrightarrow & (1,1) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,0) & \longrightarrow & (1,0) & \longrightarrow & \cdots \end{array}$$

Vertical arrows represent the second arc type; include the wrap arrows from $i = m-1$ to $i = 0$ and from $j = n-1$ to $j = 0$. These rules specify every labeled vertex and arc, including the boundary identifications.

Exercise 11.31: Two concentric polygon digraphs

Let r denote the solid-arrow generator, of order n, and s the dashed-edge generator, of order 2. Label one polygon $e, r, \ldots, r^{n-1}$ and the matching vertices in the other $s, rs, \ldots, r^{n-1}s$. **(a)** The group is abelian exactly when the arrows on the two polygons run in the same direction. In that case traversing a dashed edge before or after a solid edge reaches the same vertex, so $rs = sr$; because r, s generate the group, this proves commutativity. Opposite directions instead give $sr = r^{-1}s \neq rs$ for $n \geq 3$. **(b)** In the abelian case all $2n$ elements are uniquely $r^i s^j$ with $0 \leq i < n$, $j = 0, 1$, and multiplication adds both exponents. Thus $G \cong \mathbb{Z}_n \times \mathbb{Z}_2$. **(c)** This product is cyclic exactly when $\gcd(n, 2) = 1$, that is, n is odd. **(d)** Opposite arrow directions give exactly the usual dihedral relations, and hence the $2n$-element group is D_n.

Exercise 11.32: True or false: direct products

(a) True: $(g_1, g_2) \mapsto (g_2, g_1)$ is an isomorphism.

(b) Componentwise computation is precisely the multiplication rule; the description "easy" is informal, but the intended mathematical content is true.

(c) False: $\mathbb{Z} \times \mathbb{Z}$ is a direct product of infinite groups.

(d) True: the orders of two proper nontrivial finite factors would both exceed 1 and multiply to the prime group order, which is impossible.

(e) False: every element of $\mathbb{Z}_2 \times \mathbb{Z}_4$ has order dividing 4, unlike a generator of $\mathbb{Z}_8$.

(f) False: the product has 8 elements, while S_8 has 8!; it is also abelian.

(g) False: both groups have 24 elements, but $\mathbb{Z}_3 \times \mathbb{Z}_8$ is abelian and S_4 is not.

(h) False: the identity has order 1; other orders can also be 2 or 4.

(i) False: $|\mathbb{Z}_{12} \times \mathbb{Z}_{15}| = 12 \cdot 15 = 180$.

(j) True: there are independently m choices in the first coordinate and n in the second.

Exercise 11.33: An indecomposable abelian group

Take $G = \mathbb{Z}_p$ for a prime p. Its only subgroups are $\{0\}$ and itself, so there are no proper nontrivial subgroups to use as internal factors. Alternatively, an internal product of two nontrivial finite groups would have composite order, contrary to $|G| = p$.

Exercise 11.34: Subgroups isomorphic to the full product

(a) Exactly one. The finite group $\mathbb{Z}_5 \times \mathbb{Z}_6$ has 30 elements, and a subgroup isomorphic to it must also have 30 elements, hence equal the whole group. **(b)** Infinitely many: for each positive integer k, $k\mathbb{Z} \times \mathbb{Z}$ is a subgroup of $\mathbb{Z}^2$ isomorphic to $\mathbb{Z}^2$ via $(a, b) \mapsto (ka, b)$. These subgroups are distinct. More precisely there are countably infinitely many, since any subgroup isomorphic to $\mathbb{Z}^2$ has two generators in the countable set $\mathbb{Z}^2$, and there are only countably many ordered pairs of possible generators.

Exercise 11.35: A composite-order group with no nontrivial direct decomposition

Take $\mathbb{Z}_4$. Its only proper nontrivial subgroup is $\{0, 2\}$. Any two proper nontrivial internal factors would therefore coincide and have nontrivial intersection, so cannot give an internal direct product. Equivalently the only possible nontrivial factor orders are $2, 2$, whose external product is $\mathbb{Z}_2^2$, with no element of order 4, and hence is not $\mathbb{Z}_4$.

Exercise 11.36: True or false: classification of abelian groups

(a) True: every group of prime order is cyclic.

(b) False: $\mathbb{Z}_2^2$ has prime-power order 4 and is not cyclic.

(c) False: every sum of 4's and 6's modulo 8 is even, so these generate only $\{0, 2, 4, 6\}$.

(d) True: $5 - 4 = 1$, so $\{4, 5, 6\}$ generates 1 and hence $\mathbb{Z}_8$.

(e) True: a finite group is finitely generated, so the theorem applies.

(f) False: $\mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z}_2$ both have Betti number 1, but only the latter has nontrivial torsion.

(g) True: the primary decomposition contains a 5-power cyclic factor, which has an element of order 5.

(h) False: $\mathbb{Z}_2^2$ has order divisible by 4 but no element of order 4.

(i) True: the primary decomposition contains elements of orders 2 and 3 in different primary factors; their product has order 6.

(j) True: a nonzero free $\mathbb{Z}$ factor would make the group infinite, so a finite group's Betti number is 0.

Exercise 11.37: Changing the prime in a prime-power order

The numbers are equal. For either prime, the isomorphism types are parametrized by the same partitions of the exponent r. Replacing every factor $\mathbb{Z}_{p^k}$ in a decomposition by $\mathbb{Z}_{q^k}$ gives a bijection

of the two lists of isomorphism types. Thus the count depends on r alone, not on the prime.

Exercise 11.38: Order-eight and order-four subgroups of an order-72 group

Write $G = P \times Q$ with $|P| = 8$ and $|Q| = 9$, using the primary decomposition. **(a)** There is exactly one subgroup of order 8, namely $P \times \{0\}$. An element of a group of order 8 has order a power of 2, so its projection into the order-9 group Q is zero. Thus any order-8 subgroup lies in $P \times \{0\}$ and must equal it. **(b)** The number of order-4 subgroups is not determined by $|G|$ alone. If $P = \mathbb{Z}_8$, it has exactly one order-4 subgroup. If $P = \mathbb{Z}_2^3$, it has seven planes of order 4. Taking, for example, $Q = \mathbb{Z}_9$ gives two groups of order 72 with different answers.

Exercise 11.39: The torsion subgroup

Let T consist of elements of finite order. The identity is in T. If $a^r = e$ and $b^s = e$, commutativity gives $(ab)^{rs} = a^{rs}b^{rs} = e$, so ab has finite order and belongs to T. If $a^r = e$, then $(a^{-1})^r = (a^r)^{-1} = e$, so $a^{-1} \in T$. Thus T is a subgroup. The commutativity assumption is used in expanding the power of ab; it cannot simply be omitted.

Exercise 11.40: Orders of two torsion subgroups

In a product with a $\mathbb{Z}$ coordinate, finite order forces that coordinate to be 0, because a nonzero integer has infinite additive order. All elements in the remaining finite factors are torsion. Hence the torsion subgroups are

$$\mathbb{Z}_4 \times \{0\} \times \mathbb{Z}_3 \quad \text{of order } 12,$$

$$\mathbb{Z}_{12} \times \{0\} \times \mathbb{Z}_{12} \quad \text{of order } 144.$$

Exercise 11.41: Torsion in the nonzero real numbers

If $x^n = 1$ for a real nonzero x, then $|x|^n = 1$, so $|x| = 1$ and $x = \pm 1$. Conversely 1 has order 1 and -1 has order 2. Thus the torsion subgroup is $\{1, -1\} \cong \mathbb{Z}_2$.

Exercise 11.42: Torsion in the nonzero complex numbers

A complex number has finite multiplicative order exactly when it is a root of unity. Indeed $z^n = 1$ implies $z = e^{2\pi ik/n}$ for some integer k, and any such number has a finite power equal to 1. Therefore

$$T = \{e^{2\pi iq} : q \in \mathbb{Q}\},$$

the group of all roots of unity. It is isomorphic to $\mathbb{Q}/\mathbb{Z}$ under the map $q + \mathbb{Z} \mapsto e^{2\pi iq}$.

Exercise 11.43: Torsion and free parts of a finitely generated abelian group

By the classification theorem choose an isomorphism $G \cong F \times \mathbb{Z}^r$, where F is a finite product of finite cyclic prime-power groups. An element has finite order precisely when all its $\mathbb{Z}$ coordinates are zero, so the torsion subgroup corresponds to $F \times \{0\}$. The subgroup corresponding to $\{e\} \times \mathbb{Z}^r$ is torsion

free. These subgroups intersect trivially, commute elementwise, and every element is their product; transporting them back through the isomorphism gives an internal direct-product decomposition of G. This includes the case of a trivial finite or free part.

Exercise 11.44: Torsion coefficients

(a) Since $\gcd(4, 9) = 1$, $\mathbb{Z}_4 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36}$. The only torsion coefficient is 36. **(b)** The group is $\mathbb{Z}_6 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20}$. Split into prime-power factors and sort the exponents at each prime, padding with zero exponents on the left:

p	first	second	third
2	1	2	2
3	0	1	1
5	0	0	1

Combining each column gives 2, $2^2 \cdot 3 = 12$, and $2^2 \cdot 3 \cdot 5 = 60$. Hence the torsion coefficients are $2, 12, 60$, and $2 \mid 12 \mid 60$. **(c)** For an arbitrary finite product, factor all moduli into prime powers. For each prime put the exponents in nondecreasing order and pad shorter lists with zero exponents at their left ends to obtain equal lengths. Multiply the prime powers column by column. The resulting column products divide the next ones, because every prime exponent is nondecreasing. Reordering primary factors and combining relatively prime factors proves this process preserves the isomorphism type. Discard any leading coefficient 1.

Exercise 11.45: Synopsis of the coprime-product criterion

The element $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ has order $\operatorname{lcm}(m, n)$, so when $\gcd(m, n) = 1$ it generates the entire order-mn group. If the gcd is $d > 1$, every element is killed by $mn/d < mn$, so no element can generate the whole group and the product is not cyclic.

Exercise 11.46: A direct product of abelian groups is abelian

Let $a = (a_1, \ldots, a_r)$ and $b = (b_1, \ldots, b_r)$ belong to the product. Since each factor is abelian, $a_i b_i = b_i a_i$ for every i. Therefore

$$ab = (a_1 b_1, \ldots, a_r b_r) = (b_1 a_1, \ldots, b_r a_r) = ba.$$

This is exactly commutativity of the product operation.

Exercise 11.47: Elements of order at most two

The given set is $H = \{g \in G : g^2 = e\}$. It contains e. If $a, b \in H$, commutativity gives $(ab)^2 = a^2 b^2 = e$, so $ab \in H$. Also $(a^{-1})^2 = (a^2)^{-1} = e$. Thus H is a subgroup. Any nonidentity element killed by 2 has order exactly 2, so this is precisely the set described in the question.

Exercise 11.48: Identity together with all elements of an exact order

For order 3, the answer is yes: the set is $\{g : g^3 = e\}$, and commutativity shows it is closed under products and inverses. For order 4, the answer is no. In $\mathbb{Z}_4$, 1 has order 4 but $1 + 1 = 2$ has order 2, so the set $\{0, 1, 3\}$ is not closed. In general the universal property holds exactly for $n = 1$ and for prime

n. For prime p, every nonidentity solution of $g^p = e$ has order p, so the same kernel-style closure argument works. If n is composite, choose a prime $p \mid n$ with $p < n$ and work in $\mathbb{Z}_n \times \mathbb{Z}_n$. Both

$$a = (1, 0), \qquad b = (-1, n/p)$$

have order n because their first coordinates do. But $a + b = (0, n/p)$ has order p, which is neither 1 nor n. Thus the prescribed set fails closure. This also explains the distinction between exact order n and order dividing n.

Exercise 11.49: Why commutativity is needed for involutions

In the nonabelian group S_3, the identity and three transpositions are exactly the elements of order 1 or 2. But $(1, 2)(2, 3) = (1, 2, 3)$ has order 3 and does not lie in that set. Hence it is not a subgroup, providing a counterexample when the abelian hypothesis is removed.

Exercise 11.50: The coordinate subgroups in an external product

Write the two coordinate subgroups as $\widetilde{H} = H \times \{e_K\}$ and $\widetilde{K} = \{e_H\} \times K$. **(a)** Every (h, k) equals $(h, e_K)(e_H, k)$. **(b)** Componentwise multiplication gives

$$(h, e_K)(e_H, k) = (h, k) = (e_H, k)(h, e_K),$$

so the two subgroups commute elementwise even if the individual groups are nonabelian. **(c)** An element in both coordinate subgroups has first coordinate e_H and second coordinate e_K, so the intersection is exactly $\{(e_H, e_K)\}$.

Exercise 11.51: Recognizing an internal direct product

By the first assumed condition, every $g \in G$ is hk. If $hk = h'k'$, rearrange to obtain

$$h'^{-1}h = k'k^{-1} \in H \cap K = \{e\}.$$

Thus $h = h'$ and $k = k'$, proving uniqueness. Define $\Phi : H \times K \to G$ by $\Phi(h, k) = hk$. Existence and uniqueness say it is bijective. Since every element of H commutes with every element of K,

$$\Phi(h, k)\Phi(h', k') = hkh'k' = hh'kk' = \Phi(hh', kk').$$

It is therefore a homomorphism and an isomorphism, making precise the claimed structural identity.

Exercise 11.52: Detecting a noncyclic finite abelian group

Suppose first G contains $\mathbb{Z}_p^2$. This subgroup is not cyclic: every nonidentity element has order p, while the group has order p^2. Every subgroup of a cyclic group is cyclic, so G cannot be cyclic. Conversely, decompose finite abelian G into its prime-power cyclic factors. If each prime occurred in at most one factor, these factor orders would be pairwise coprime, and their product would be cyclic. Since G is not cyclic, some prime p occurs in at least two factors, say $\mathbb{Z}_{p^a}$ and $\mathbb{Z}_{p^b}$. Their order-p subgroups generated by p^{a-1} and p^{b-1} have direct product $\mathbb{Z}_p^2$, which embeds in G. This proves the equivalence.

Exercise 11.53: Element orders in a prime-power group

If $|G| = p^r$, then for any $g \in G$, Lagrange's theorem says $\operatorname{ord}(g)$ divides p^r. The positive divisors are exactly p^s for $0 \leq s \leq r$, so every element order is a power of p (including $p^0 = 1$). This proof did not use commutativity; therefore the abelian hypothesis can be dropped.

Exercise 11.54: Cancellation for finitely generated abelian groups

By classification, a finitely generated abelian group is specified up to isomorphism by its free rank and, for every prime p and exponent $j \geq 1$, the multiplicity of the factor $\mathbb{Z}_{p^j}$. In a direct product these ranks and multiplicities add. The isomorphism $G \times K \cong H \times K$ therefore gives

$$r_G + r_K = r_H + r_K, \qquad m_G(p, j) + m_K(p, j) = m_H(p, j) + m_K(p, j).$$

All quantities are finite nonnegative integers, since the groups are finitely generated. Cancelling the common terms yields equal free ranks and all equal prime-power multiplicities for G, H. The uniqueness part of the theorem then gives $G \cong H$.